# Benchmark Hygiene Reveals Label Circularity and Ground-Truth Contamination in Peptide-MHC-I Binding Prediction

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## Abstract

Peptide-MHC-I binding predictors are routinely trained on computationally labelled data, yet benchmark scores that merely reproduce the labeler's decision boundary are rarely distinguished from true binding accuracy. We quantify a failure mode — **label circularity** — and distil a benchmark-hygiene checklist to detect it. Re-analysing a 5,190-peptide HLA-A*02:01 benchmark under controlled conditions dissolved three apparent findings — per-allele superiority over a pan model, "label quality dominates architecture", and "simple beats complex" — each traced to a distinct confound (a 10.6× data-size artifact, a degenerate null, and stale checkpoints). A zero-parameter anchor rule underperformed the trained model on non-canonical binders, which reached ~0.65 sensitivity across four predictors at unmatched operating points, including BigMHC. A 6,524-affinity benchmark (AUC 0.942) and a post-2022 temporal holdout bounded label self-consistency and generalisation. Code, models, and data are available at github.com/woodhaha/peptide-mhc-benchmark.

## 1 Introduction

MHC class I molecules present 8–11-mer peptides to CD8+ T cells, and accurate prediction of peptide–MHC-I binding underpins epitope discovery for vaccine design and cancer immunotherapy [1,2]. For HLA-A*02:01 — the most prevalent class I allele globally — binding is dominated by two primary anchor positions: position 2 (p2, hydrophobic: Leu/Met/Ile/Val) and position 9 (p9, C-terminal, Val/Leu preferred) [3,4]. Computational predictors have evolved from position-specific scoring matrices to neural networks and pan-specific tools such as NetMHCpan [5], MHCflurry [6], BigMHC [7], and MUNIS [8], increasingly using protein language model embeddings [9].

Several gaps remain. A particularly under-examined one is **label circularity**: because >55% of IEDB entries are computationally rather than experimentally labelled [10], a predictor's apparent accuracy can track the decision boundary of whichever labeler trained it rather than true binding. The impact of training-label quality is seldom quantified, and no reusable checklist exists for detecting such confounds. Head-to-head architecture comparisons under fully controlled conditions (identical training data, encoding, evaluation) are uncommon. Whether a single pan-specific model can replace per-allele specialists, and how hard non-canonical binders are across alleles, remains unsettled. And the translation of binding prediction to real-world epitope discovery is rarely demonstrated in a unified, reproducible pipeline.

To isolate the effects of architecture choice, label quality, and encoding, we present a controlled benchmark. We compare six architectures under multi-seed uncertainty and find a modest but measurable architecture effect, confirm that random labels fall to chance, and evaluate ESM-2 embeddings against BLOSUM62 with a mean-pooled ablation. We further validate against 42 A*02:01-verified IEDB epitopes, scan proteins and cancer mutations as a translational application, and extend to five HLA-I alleles and a 15-allele panel to test whether a single rebalanced pan model matches per-allele specialists and to quantify how hard non-canonical binders are for four independent sequence predictors.

## 2 Methods

### 2.1 Training data and labels

100,000 random 9-mer peptides were generated by uniform sampling of the 20 standard amino acids. Each was labelled with MHCflurry

2.2.1 against HLA-A*02:01 and assigned to Strong Binder (SB, percentile rank <0.5%), Weak Binder (WB, 0.5%≤rank<2.0%), or Non-Binder (NB, rank ≥2.0%) following the netMHCpan convention [5]. The resulting distribution (1.7% SB, 5.1% WB, 93.2% NB) was balanced by downsampling to 1,730 per class (n=5,190), partitioned 90/10 into training (n=4,671) and held-out test (n=519) sets with stratified sampling (seed 42). Two comparator labelling strategies were implemented: a PSSM encoding crystallographic A*02:01 anchor preferences, and random synthetic labels as a baseline control. A held-out benchmark of 42 experimentally validated IEDB epitopes (HLA-A*02:01-restricted, with positive binding/T-cell assay evidence, re-verified against the IEDB on 2026-10-23) plus 20 homopolymer negatives (≤2 unique amino acids) was curated from the IEDB (query-api.iedb.org); label-circularity overlap was determined by sequence-identity match against the IEDB entries in MHCflurry's training corpus.

### 2.2 Peptide encoding

Each 9-mer was encoded with the BLOSUM62 matrix [11], min–max normalized, producing a 9×20 matrix flattened to 180 dimensions for fully connected networks. ESM-2 embeddings [12] were extracted at t6 (2,880-dim per-peptide; 320 per position × 9) and t12 (4,320-dim; 480 per position × 9) depths — the two smallest ESM-2 checkpoints (8M and 35M parameters); a mean-pooled t6 variant (320-dim) served as an ablation control destroying positional information.

### 2.3 Model architectures

Six classifiers were implemented: (i) FFN [13]: 180→90→3, 49K params; (ii) Deep FFN: 180→360→180→90→45→3 with batch normalisation and dropout, 152K params; (iii) CNN: Conv2D(32,3×3) + FFN body, 1.05M params; (iv) LSTM(64)→dense(32)→3, 24K params; (v) ResNet-style stem + 3 residual blocks, 325K params; (vi) Random Forest (100 trees). For ESM-2 embeddings an FFN classifier with a 256→128→64 stack was trained. All networks used cross-entropy loss with Adam or RMSprop (lr 0.001), 150 epochs, batch size 50, 20% internal validation split, early stopping (patience 10), and learning-rate reduction on plateau. Five-fold stratified cross-validation was performed for the FFN. The held-out test set (519 peptides) was disjoint from training and used for all reported metrics.

### 2.4 Protein scanning, mutation analysis, and structure

Ten therapeutically relevant proteins (MART-1, gp100, tyrosinase, NY-ESO-1, WT1, p53, KRAS, influenza M1, CMV pp65, SARS-CoV-2 spike

RBD) were scanned with a sliding 9-mer window (stride 1; 3,536 windows). Sixteen recurrent hotspot mutations (7 p53, 9 KRAS) were analyzed by comparing wild-type and mutant binding scores; mutations were classified as Created/Enhanced/Destroyed/Unchanged. The KRAS G12V neoepitope YKLVVVGAV carries a non-canonical lysine at P2.

### 2.5 Feature extraction

Learned features were extracted from the penultimate dense layer via truncated models. Hierarchical clustering (Ward's method, Euclidean distance) was applied to representations of top epitope candidates; feature–binding-score correlation was assessed by Pearson's r with a 1,000-iteration permutation test to control for chance correlations. t-SNE visualisation of the 45-dimensional penultimate layer was used to inspect class separation, and integrated gradients identified position-specific anchor contributions.

### 2.6 Multi-allele extension

**2.6.1 Multi-allele data curation** Experimentally confirmed class I binders were curated from the IEDB [14] for five common alleles (HLA-A*02:01, A*01:01, A*03:01, B*07:02, B*44:03), yielding n = 14,488 positives (canonical 11,225; Type A 151; Type B 3,112). A 15-allele panel (pan16: 11 additional alleles — A*11:01, A*24:02, B*08:01, B*15:01, B*27:05, B*35:01, B*40:01, B*49:01, B*51:01, B*57:01) was curated for generalisation testing. Protein-background negatives were sampled from natural proteomes to avoid the uniform-random decoys used in earlier evaluations. IEDB data were queried live from query-api.iedb.org (accessed 2026-08-02); the 15-allele and temporal-holdout estimates are therefore snapshot-dependent and drift slightly as the IEDB grows.

**2.6.2 Type A/B classification** Type A (tolerated weak anchors) and Type B (energy-compensated non-canonical) were defined by whether the peptide violates the allele's canonical anchor motif at p2/p9. Type A carries broad-only tolerated anchors (residues that appear in binding motifs but are non-canonical); Type B carries non-canonical residues at anchor positions. The two classes have different biological origins and, as we show, different prediction difficulty.

**2.6.3 Pan model and per-allele comparators** The pan-specific DFFN uses BLOSUM62 encoding [11] through a 360–180–90–45 hid-

den stack with batch normalisation and dropout. Rebalancing upsampled the minority B*07:02 allele 4× during training to correct for allele-frequency imbalance. Five seeds (0, 1, 42, 123, 2024) were used; per-seed variation arises from data-sampling RNG. Per-allele models (one per allele) and a pan16 model (15 alleles, per-allele minimum 2,000 upsampling) were trained with the same recipe.

**2.6.4 Comparator predictors** MHCflurry 2.2.1 [6] served as an independent sequence predictor (affinity threshold 500 nM). MUNIS [8] — a second sequence-based predictor using ESM-2 embeddings [9] — served as an independent sequence comparator, run on the n = 3,112 Type B population plus canonical controls.

**2.6.5 Temporal holdout** To address label circularity [10], IEDB epitopes published after 2022 (post-2022, unseen by MHCflurry's ~pre-2021 training data) were compared against pre-2021 (in-distribution) epitopes at sensitivity threshold 0.95.

**2.6.6 Evaluation** Sensitivity was computed with protein-background negatives and fixed thresholds (0.95 for the pan model; 500 nM for MHCflurry; 5th percentile of canonical scores for MUNIS and BigMHC). Wilson 95% confidence intervals are reported for pooled estimates.

## 3 Results

### 3.1 Training data and architecture comparison

MHCflurry labelling of 100,000 random 9-mers yielded 1.7% SB, 5.1% WB, 93.2% NB; after balancing, 5,190 peptides (1,730/class). Across five training seeds, the CNN achieved the highest accuracy (91.9% ± 0.9%; Table 1, Figure 1), modestly ahead of the Deep FFN (90.6% ± 1.2%), the FFN (89.8% ± 1.1%), and the LSTM (89.7% ± 0.9%) — the top four architectures within ~2 percentage points, a range within the combined seed-plus-binomial uncertainty (n=519; 95% CI CNN [0.889, 0.949] vs Deep FFN [0.872, 0.940]) — while ResNet (87.6% ± 1.8%) and Random Forest (79.7% ± 0.4%) were clearly worse. Two external predictors frame this comparison (Table 1): MHCflurry 2.2.1 itself scores 100% on the labels it produced — an expected ceiling, not evidence of superiority — while the independent state-of-the-art predictor NetMHCpan-4.1, which was not trained on these labels, agrees with them on only 62.24% (323/519). The 8.1-point gap between the labeler (100%) and the best finite-sample approximation (91.9% for the CNN) is information loss from finite samples; the far larger 28-point gap to NetMHCpan (62.24%) reflects, in unknown proportion,

label-definition divergence and label circularity (see Discussion). The WB class was consistently the most difficult (F1 0.752–0.880).

**Table 1.** Model performance on the MHCflurry-labelled test set (mean ± SD across five training seeds, n=519).

| Model | Accuracy (%) | Parameters |
|---|---|---|
| CNN | **91.9 ± 0.9** | 1,053,863 |
| Deep FFN | 90.6 ± 1.2 | 152,823 |
| FFN (Nielsen) | 89.8 ± 1.1 | 49,143 |
| LSTM | 89.7 ± 0.9 | 23,939 |
| ResNet | 87.6 ± 1.8 | 324,931 |
| Random Forest | 79.7 ± 0.4 | – |
| MHCflurry 2.2.1 (labeler) | 100 | — |
| NetMHCpan-4.1 (independent SOTA) | 62.24 | — |

† Accuracy is agreement with the MHCflurry test labels (n=519). MHCflurry scores 100% because it produced the labels — an expected ceiling, not evidence of superiority. NetMHCpan-4.1, which was not trained on these labels, agrees with them on only 62.24%, quantifying label circularity (see Discussion).

The modest spread among the top four architectures reflects the positional nature of peptide–MHC binding: anchor positions p2/p9 dominate, so architectures that preserve positional information perform comparably. ResNet and the Random Forest were clearly worse, consistent with the absence of translation invariance in a 9×20 encoding and, for the Random Forest, the high dimensionality of the BLOSUM62 encoding.

### 3.2 Label-source controls

Label source, not architecture, sets the practical ceiling of what a model can learn (Table 2, Figure 2). Training on MHCflurry labels reached 90.6% accuracy, whereas a deterministic PSSM label source reached 82.6% (on a separate 15,000-peptide pool, so not a clean head-to-head comparison) and an i.i.d. random-label null fell to chance (34.5 ± 1.7%, close to the 33.3% expected for three balanced classes). The random-label result confirms the models learn genuine signal rather than noise; the PSSM–MHCflurry gap, though confounded by pool size, suggests MHCflurry's labels are at least as learnable as a hand-crafted scoring matrix.

**Table 2.** Deep FFN performance across label sources (accuracy, mean ± SD across five seeds where noted).

| Label source | Accuracy (%) | Notes |
|---|---|---|
| DFFN on MHCflurry labels | 90.6 ± 1.2 | 5,190 peptides, 5 seeds |
| PSSM (biophysics) | 82.6 | separate 15,000-peptide pool, single seed |
| Random (i.i.d.) | 34.5 ± 1.7 | chance-level null, 5 seeds |

### 3.3 Encoding

On the FFN (the architecture used for all encoding comparisons), BLOSUM62 generalized stably under cross-validation (92.2 ± 0.4%; per-fold 92.2/92.4/92.1/92.7/91.6%). ESM-2 t6 per-position embeddings scored 93.3% on a single split but collapsed to 82.2 ± 1.0% under CV — the 2,880-dimensional features overfit the 4,671-sample training set (Table 3, Figure 7). Mean-pooled t6 embeddings collapsed to 65.9%, confirming that positional information is essential. The t12 variant (4,320-dim) overfit similarly (90.9–91.5%). The single-split 93.3% result reflects a favourable partition rather than a genuine encoding advantage at this training scale. ESM-2's representational advantage materialises only at large scale: the 650M checkpoint (11,520-dim per-peptide) reached 92.4 ± 0.5% under CV, matching BLOSUM62 (92.2 ± 0.4%) but at 64× the input dimensionality. At the data scale typical of single-allele studies, compact biologically motivated encodings such as BLOSUM62 therefore remain the more cost-effective choice, while the small ESM-2 checkpoints overfit.

**Table 3.** Classification accuracy by encoding strategy.

| Encoding | Dim | Accuracy (%) | Notes |
|---|---|---|---|
| BLOSUM62 | 180 | 92.2 ± 0.4 (CV) | Stable CV generalisation |
| ESM-2 t6 mean pool | 320 | 65.9 | Positional info destroyed |
| ESM-2 t12 per-pos | 4,320 | 90.9 | Overfitting |
| ESM-2 t12 + L2 | 4,320 | 91.5 | Partially mitigated |
| ESM-2 t6 per-pos | 2,880 | 93.3 (82.2 ± 1.0 CV) | Single-split optimistic |
| ESM-2 t33 (650M) per-pos | 11,520 | 92.4 ± 0.5 (CV) | Matches BLOSUM62 at 64× dim |

### 3.4 Held-out validation and translation

The Deep FFN achieved 97.6% sensitivity (41/42; Wilson 95% CI 87.7–99.6%), 75.0% specificity (15/20), ROC AUC 0.970 against IEDB epitopes (Table 5, Figure 3). Independent re-verification against the IEDB (query-api.iedb.org, accessed 2026-10-23) excluded seven peptides lacking A*02:01 restriction: LLFDRFENL and LLHHAFDSL (no IEDB records), YVKQNTLKL (HLA-B*07:02), ILRGSVAHK (HLA-A*03:01; previously mis-attributed to B*08:01), QYDPVAALF and CMTWNQMNL (HLA-A*24:02), and VIYQYMDDL (no positive A*02:01 record). The single remaining false negative, TLGIVCPIC (HPV16 E7), carries a non-canonical p9 anchor (C). An independent predictor, NetMHCpan-4.1 (queried through the IEDB web API, which does not pin a reproducible tool version), achieved 83.3% sensitivity (40/48) on the original 48-peptide set (not re-evaluated on the corrected 42-peptide set). This gap partly reflects label circularity (most of the epitopes overlap MHCflurry's training corpus), but NetMHCpan's pan-specific training on MS-elution-based labels makes this an indicative rather than clean comparison; the more direct circularity evidence is the 62.24% label agreement (323/519 test peptides) reported in the Discussion. To contextualise the benchmark, we include a zero-parameter anchor-checker baseline (a rule classifying any 9-mer with canonical p2/p9 residues as a binder): it achieves sensitivity 0.786 (33/42) and specificity 0.900 (18/20), below the trained model (0.976), because nine genuine binders carry non-canonical p2/p9 anchors. The trained model recovers eight of these nine — the single residual miss, TLGIVCPIC (p9 = C), defines the hard non-canonical tail — confirming that the informative signal lies beyond the anchor motif. The five false positives were homopolymers (≤2 unique amino acids) with canonical anchors — biologically implausible binders whose low sequence complexity the classifier fails to penalise. A domain-knowledge filter (≤2 unique amino acids → non-binder) removed all five, but we report 75.0% as the primary, unfiltered specificity to avoid post-hoc inflation. To break this circularity, we additionally scored the Deep FFN against a large label-independent set of experimentally measured affinities (IEDB binding-affinity corpus, accessed 2026-08-02) — 3,438 unique 9-mers with measured IC50 <500 nM and 3,086 with IC50 >5,000 nM (the 500–5,000 nM grey zone was excluded to yield unambiguous binders and non-binders), restricted to direct binding assays (competitive fluorescence/radioactivity) and excluding MS-elution-derived, predictor-dependent entries. The MHCflurry-trained Deep FFN recognised 94.4% of the experimental binders and rejected 74.5% of the non-binders (threshold 0.5; ROC AUC 0.942); the rebalanced per-allele model reached 89.8% sensitivity and 89.1% specificity (AUC 0.945). This benchmark validates the quality of the training labels — that MHCflurry's computational

labels agree with experimentally measured binding — rather than generalisation to unseen peptides, since the peptides are drawn from the same IEDB binding-affinity corpus on which the labeler was trained; generalisation to unseen peptides is instead established by the temporal holdout (§3.11). Protein scanning of 10 proteins (3,536 windows) recovered 10/11 known epitopes (91%; Figure 4), including immunodominant GILGFVFTL (M1), NLVPMVATV (pp65), and SLLMWITQC (NY-ESO-1); the single miss was the MART-1 clinical variant ELAGIGILTV, whose P2 leucine is a non-natural anchor-modified residue. Mutation scanning of 16 hotspot mutations identified 7 epitope-altering events (Table 4, Figure 5). p53 R248W created the neoepitope MNWRPILTI (R→W at p3 introduces a bulky hydrophobic residue improving groove fit); KRAS G12V created YKLVVVGAV via G→V at p9 creating a canonical C-terminal anchor. p53 R248W is among the most frequent TP53 missense mutations across cancers (~3.5% of somatic TP53 mutations, IARC), and mutation-specific T-cell responses have been documented in patients and in HLA-A2/DRB1-transgenic mice [15,16]; KRAS G12C enhanced an existing epitope (WB→SB), suggesting synergy with pharmacological G12C inhibition. Three mutations (KRAS G13D, KRAS A146T, p53 R249S) destroyed existing epitopes, representing potential immune-evasion mechanisms.

**Table 4.** Epitope-altering cancer hotspot mutations.

| Mutation | WT peptide | Mutant peptide | Effect | Δ score |
|---|---|---|---|---|
| p53 R248W | MNRRPILTI | MNWRPILTI | CREATED | +0.41 |
| KRAS G12V | YKLVVVGAG | YKLVVVGAV | CREATED | +0.48 |
| KRAS G12V | LVVVGAGGV | LVVVGAVGV | ENHANCED | +0.31 |
| KRAS G12C | LVVVGAGGV | LVVVGACGV | ENHANCED | +0.31 |
| KRAS G13D | GVGKSALTI | DVDKSALTI | DESTROYED | −0.47 |
| KRAS A146T | GIPFIETSA | GIPFIETST | DESTROYED | −0.42 |
| p53 R249S | GMNRRPILT | GMNRSPILT | DESTROYED | −0.16 |

**Table 5.** IEDB benchmark performance.

| Metric | Value |
|---|---|
| Sensitivity | 97.6% (41/42) |
| NetMHCpan-4.1 sensitivity | 83.3% (40/48)* |
| Specificity | 75.0% (15/20) |
| Precision | 89.1% |
| F1 | 0.932 |
| ROC AUC | 0.970 |

*NetMHCpan-4.1 not re-evaluated on the corrected 42-epitope set.

### 3.5 A rebalanced pan model matches per-allele specialists on four of five alleles

With protein-background negatives and a fixed threshold (n=25 non-canonical binders per allele, so these point estimates carry wide confidence intervals), per-allele models reached non-canonical sensitivity A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*07:02 0.96, B*44:03 0.96 (Table 6; Figure 8). The rebalanced pan model matched this on four of five alleles (A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*44:03 1.00), but B*07:02 was the clear exception: the pan model dropped to 0.60 versus 0.96 per-allele (0.64 in the 15-allele pan16 panel; Wilson 95% CIs [0.41, 0.77] vs [0.81, 0.99], non-overlapping), consistent with over-anchoring to B*07:02's distinctive P2-Proline — a reproducible limitation rather than a fundamental pan-model deficiency. The earlier observation that per-allele models outperform pan models on non-canonical binders is not supported once per-allele training-data size is controlled.

**Table 6. Non-canonical sensitivity by allele (protein-background negatives, n=25 per allele, @0.95).**

| Allele | Per-allele | Pan |
|---|---|---|
| A*02:01 | 0.84 | 0.84 |
| A*01:01 | 0.88 | 0.88 |
| A*03:01 | 1.00 | 1.00 |
| B*07:02 | 0.96 | **0.60** |
| B*44:03 | 0.96 | 1.00 |

### 3.6 Canonical binders are recognized at ~0.98 by every predictor

On n = 11,225 canonical positives, all four predictors were saturated: DFFN 0.979, MHCflurry 0.986, MUNIS 0.950, BigMHC 0.950 (Table 7). The canonical binder problem is effectively solved by all current methods.

### 3.7 Non-canonical (Type B) binders plateau at ~0.65 across predictor classes

On n = 3,112 Type B positives, sensitivity was DFFN 0.607, MHCflurry 0.668, MUNIS 0.647, and BigMHC 0.660 (Table 7; Figure 9). The four values fall within a narrow band (0.607–0.668), far below the ~0.98

canonical sensitivity, and the attention-based BigMHC does not lift the ceiling either. This convergence is, however, partly a threshold artifact: the predictors operate at different points (DFFN at a fixed 0.95 score, MHCflurry at 500 nM, MUNIS and BigMHC at the 5th percentile of canonical scores), and when aligned to matched canonical sensitivity (0.95) the Type B sensitivities diverge — DFFN 0.454, MHCflurry 0.497, MUNIS 0.647, BigMHC 0.653. The apparent plateau therefore overstates the degree to which the non-canonical ceiling is shared across architectures.

**Table 7. Predictor sensitivity by binder class (pooled, 5 alleles).**

| Predictor | Canonical (n=11,225) | Type A (n=151) | Type B (n=3,112) |
|---|---|---|---|
| DFFN (sequence) | 0.979 | 0.477 | 0.607 |
| MHCflurry (sequence) | 0.986 | 0.669 | 0.668 |
| MUNIS (sequence) | 0.950 | — | 0.647 |
| BigMHC (sequence) | 0.950 | 0.600 | 0.660 |

### 3.8 Type A (tolerated weak anchors) is under-represented in training

On n = 151 Type A positives, DFFN achieved 0.477 and MHCflurry 0.669 (MUNIS not evaluated). Broad-motif retraining did not recover Type A (Table 8); the low sensitivity reflects under-representation of these heterogeneous, weak binders in the training data (independent predictors MHCflurry 0.669 and BigMHC 0.600 outperform DFFN 0.477 on the same set).

**Table 8. Broad-motif retraining does not recover Type A.**

| Allele | Type A before | Type A after | Type B after |
|---|---|---|---|
| A*03:01 | — | 0.634 | — |
| A*02:01 | 0.08 | 0.24 | — |
| B*07:02 | — | — | 0.341 (from 0.680) |

### 3.9 The extreme tail is prediction-resistant, cross-predictor

The single-allele false negative (TLGIVCPIC) and the KRAS G12V P2-Lys epitope YKLVVVGAV were scored as non-binders by all three predictors (MHCflurry 3.8–13.8 μM; DFFN 0.000–0.254; MUNIS 3/4 low), despite experimental CTL evidence — albeit the weakest of three mutants screened [17].

### 3.10 Scaling: one model approximates a 15-allele panel

The rebalanced pan16 reached pooled non-canonical sensitivity 0.819 (307/375, @0.95), with 12/15 alleles at 0.76–1.0 (Table 9; Figure 10). The weak tail — B*27:05 (0.08), B*51:01 (0.52), B*07:02 (0.64), A*24:02 (0.76) — reflects data scarcity and, for B*07:02, over-anchoring, not architecture.

**Table 9. Pan16 non-canonical sensitivity by allele (15-allele panel, @0.95).**

| Allele | Sens | Allele | Sens |
|---|---|---|---|
| A*02:01 | 0.88 | A*11:01 | 1.00 |
| A*01:01 | 0.80 | A*24:02 | 0.76 |
| A*03:01 | 1.00 | B*08:01 | 0.88 |
| B*07:02 | 0.64 | B*15:01 | 0.96 |
| B*44:03 | 1.00 | B*27:05 | 0.08 |
| | | B*35:01 | 0.84 |
| | | B*40:01 | 1.00 |
| | | B*49:01 | 1.00 |
| | | B*51:01 | 0.52 |
| | | B*57:01 | 0.92 |
| **Pooled** | **0.819** | | |

### 3.11 Temporal holdout shows no detectable generalisation penalty

Post-2022 epitopes (unseen by the labeler) were recognized at sens@0.95: A*02:01 0.824, A*03:01 0.936, B*07:02 0.973 — preserved versus pre-2021 (0.879/0.905/0.898; Table 10, Figure 11), with no systematic drop.

**Table 10. Temporal holdout: pre-2021 vs. post-2022 sensitivity (@0.95).**

| Allele | Pre-2021 (in-distribution) | Post-2022 (unseen) |
|---|---|---|
| A*02:01 | 0.879 | 0.824 |
| A*03:01 | 0.905 | 0.936 |
| B*07:02 | 0.898 | 0.973 |

### 3.12 Learned feature representations

Feature extraction from the Deep FFN penultimate dense layer revealed biologically meaningful structure (Figure 6). t-SNE showed

partial class separation: SB peptides clustered distinctly (91% within the SB cluster), while WB peptides formed a transition zone between SB and NB — consistent with the 50–500 nM affinity boundary. Integrated-gradients identified position-specific anchor contributions (p9 dominated SB, p2/p3 WB, NB driven by absent anchors). Hierarchical clustering separated validated epitopes, with false negatives forming a distinct branch. Five features showed Pearson $r > 0.7$ with binding score; a 1,000-iteration permutation test (≈0.3 expected by chance) confirmed genuine binding-relevant representations.

## 4 Discussion

### Methodological warnings

Three apparent findings in this project dissolved under controlled re-analysis, each illustrating how benchmark composition can manufacture false conclusions. First, an apparent per-allele superiority over a pan model was a 10.6× training-data-size artifact, not biology. Second, a claimed 'label quality dominates architecture' result collapsed once the degenerate random-label null (which performed at chance, ~33%) was removed — the realistic label effect is smaller than the architecture range. Third, a reported 'simple architecture beats complex ones' ranking was an artifact of stale model checkpoints: re-training with fixed seeds showed a CNN instead ranked above the Deep FFN (91.9% vs 90.6%, higher in all five seeds). We recommend that peptide-MHC benchmarks report multi-seed estimates with confidence intervals, protein-background negatives, frozen data snapshots, and a state-of-the-art comparator such as NetMHCpan, to avoid these failure modes. The same failure modes recur beyond peptide-MHC: data leakage from protein language-model pretraining distorts downstream task performance whenever train/test splits are not pretraining-aware [18], and leakage is now recognised as a field-wide reproducibility problem affecting hundreds of papers across seventeen fields [19]. Dedicated guidance now exists for avoiding leakage in biological machine learning [20] and for reporting clinical prediction models [21]; a recent audit of biomolecular benchmarks found cross-split contamination and label conflicts pervasive across the field [22]; Box 1 instantiates these general principles for peptide-MHC benchmarking specifically.

**Box 1. Benchmark hygiene recommendations.** The three artifacts translate into reusable practices for immunoinformatics and machine-learning benchmarks: (1) equalise training-data size across model arms and report per-allele sample counts; (2) use a true i.i.d. random-label null (chance ≈ 33% for three balanced classes) rather than a degenerate control; (3) freeze data snapshots, model check-

points, and label versions, and report multi-seed mean ± SD with confidence intervals; (4) report label provenance and break label circularity with a temporal holdout and an independent state-of-the-art comparator (NetMHCpan); (5) report performance under the natural class prior, not only a class-balanced prior, since a class-balanced benchmark overstates real-world utility. Each failure mode reproduced a false conclusion that was caught only by controlled re-analysis.

Architecture choice has a modest but measurable effect on peptide-MHC binding prediction. Across five seeds, a CNN (91.9%) ranked above a Deep FFN (90.6%) by ~1.3 percentage points (higher in all five seeds; a small margin within seed-to-seed variability), while a Random Forest was clearly worse (79.7%), for a maximum inter-architecture range of ~12 percentage points. The top four architectures fell within ~2 percentage points, indicating that data quality and the decision boundary matter more than architectural complexity for this task. Random-label controls performed at chance (~33%), confirming that the observed performance reflects genuine signal rather than noise overfitting.

A PSSM-derived label source yielded lower accuracy than MHCflurry labels (82.6% vs 90.6%, though on a separate peptide pool and therefore not a clean comparison). This does not support the hypothesis that deterministic biophysical labels are inherently superior to ML-derived labels; rather, it suggests that MHCflurry's labels are at least as learnable as a simple PSSM. We emphasise that training on any computational label — including MHCflurry — re-deposits predictor bias into downstream models, and recommend that benchmarks report label provenance and the WB-class F1 as a standard diagnostic.

An independent predictor, NetMHCpan-4.1, agreed with the MHCflurry labels on only 62.24% (323/519) of the held-out test peptides, versus 90.6% for a model trained on those labels. This ~28-point gap quantifies label circularity: reported accuracies measure fidelity to a particular labeler rather than true binding.

This circularity does not imply the training labels are uninformative: the MHCflurry-trained model retains ROC AUC 0.942 on 6,524 experimentally measured affinities (§3.4), confirming the labels carry genuine binding signal. We therefore distinguish two orthogonal properties of a labelling scheme — its **absolute predictive fidelity** (agreement with experimental binding, which is high) and its **cross-labeler transferability** (agreement with an independent predictor, 62.24%, which is low). The former determines how much signal a downstream model can learn; the latter determines how much of a reported accuracy reflects the idiosyncratic decision boundary of one labeler rather than ground truth. Benchmarks that report accuracy only against a single computational labeler conflate the two and overstate generali-

sation.

BLOSUM62 proved more robust than ESM-2 at single-allele scales: the 2,880-dimensional per-position embeddings overfit the 4,671-sample training set, and the single-split 93.3% reflects a favourable partition rather than a genuine advantage. The shallower 6-layer ESM-2 variant outperformed the deeper 12-layer model, and a PSSM label source (82.6%) underperformed MHCflurry labels. The language-model route may still be viable with continued pre-training on HLA-associated peptides [23].

The multi-allele extension shows the recipe is portable: one rebalanced pan model matches per-allele specialists on four of five alleles (B*07:02 excepted) and reaches 0.819 pooled sensitivity on 15 alleles, approximating NetMHCpan's practical value with a simple DFFN. This is consistent with the pan-specific paradigm established by mono-allelic immunopeptidomics [24], now exemplified by MixMHCpred v3.0's pan-allele model, which outperforms state-of-the-art MHC-I ligand predictors across alleles and species [25]. Yet non-canonical (motif-failing) binders plateau at ~0.65 sensitivity across four sequence-based predictors: our Deep FFN (0.607), MHCflurry (0.668), the ESM-2-based MUNIS (0.647), and BigMHC (0.660) fall within a narrow band (0.61–0.67), far below the ~0.98 canonical sensitivity. This convergence is partly a threshold artifact — the predictors operate at unmatched points, and aligned to matched canonical sensitivity the values diverge (DFFN 0.454, MHCflurry 0.497, MUNIS 0.647, BigMHC 0.653) — so the shared ceiling is weaker than the raw band suggests, though it remains far below canonical performance for every predictor. The energy-compensated tail is prediction-resistant to all four. Canonical binding is solved across methods; the non-canonical tail is where current predictors still fall short. This frontier extends beyond anchor-motif violations to non-coding, fusion-derived, and post-translationally modified peptides, for which current predictors remain under-powered [26]. A new wave of transformer- and PLM-based predictors continues to improve canonical prediction [27,28], but whether any of them lifts the non-canonical plateau remains to be seen.

### Translational implications

p53 R248W, which created `MNWRPILTI`, is among the most frequent TP53 missense mutations (~3.5% of somatic mutations, IARC); its vaccine potential requires experimental confirmation (IFN-γ ELISpot). KRAS G12C, which enhanced an existing epitope (WB→SB), is targeted by sotorasib and adagrasib, suggesting combination immunotherapy. We note that binding prediction alone does not establish immuno-

genicity: peptide–MHC stability predicts immunogenicity better than affinity [29,30], and high-throughput thermostability profiling now quantifies this at immunopeptidome scale [31,32].

### Limitations

Training labels are MHCflurry-derived rather than experimental affinities, so our single-allele benchmark is itself subject to the label circularity we describe; we quantify this by cross-referencing the 42 A*02:01-verified IEDB epitopes against MHCflurry's training corpus; the large majority overlapped, and the few non-overlapping epitopes were nonetheless correctly classified, providing limited evidence of generalisation beyond the training distribution. Similarly, the temporal holdout (§3.11) shows that post-2022 epitopes are recognised without a detectable drop, but the evaluated model's training snapshot was not time-truncated to pre-2022, so this result bounds labeler-level circularity rather than formally excluding leakage of post-2022 peptides into the model's training data. The single-allele design was deliberate — holding the allele constant isolates label-source variation from allele-driven confounds — but the multi-allele extension addresses the generalisability question directly. The multi-allele estimates rest on IEDB-derived positives; experimental (non-IEDB) validation of the extreme tail is outstanding, and a pMHC-fine-tuned structure model (beyond generic AlphaFold-Multimer) might lift the Type B plateau. All translational predictions and the cancer neoepitope candidates require experimental confirmation (e.g., IFN-γ ELISpot or tetramer staining). Antigen-processing steps (proteasomal cleavage, TAP transport) are not modelled.

## 5 Conclusion

Architecture choice has a modest effect on peptide-MHC binding prediction: a CNN ranked above the Deep FFN in all five seeds; the top four architectures fell within ~2 percentage points, and a Random Forest was clearly worse. Random-label controls confirmed chance-level performance, ruling out noise overfitting. Canonical binding is largely captured by the anchor motif, but a zero-parameter anchor rule (sensitivity 0.79) underperforms the trained model (0.98), which recovers non-canonical binders the rule misses — while non-canonical binders reach ~0.65 sensitivity across four sequence-based predictors at unmatched operating points, marking the unresolved part of the problem. Beyond peptide-MHC, the three artifacts we corrected — data-size imbalance, a degenerate null, and stale checkpoints — illustrate a general benchmark-composition failure mode that reproduces false conclusions across machine-learning benchmarks; the checklist

in Box 1 is offered as a reusable safeguard. Experimental validation of the non-canonical tail is the decisive next step.

## Figure legends

**Figure 1.** Model performance comparison across six architectures on the MHCflurry-labelled test set (accuracy, 5-seed mean ± SD).

**Figure 2.** Architecture comparison (Table 1) and label-source controls: MHCflurry (90.6%), PSSM (82.6%), random labels at chance (~33%).

**Figure 3.** IEDB benchmark: ROC curve (AUC 0.970), confusion matrix, per-epitope scores.

**Figure 4.** Protein epitope scanning: SB/WB/NB distribution across 10 proteins and top epitope candidates.

**Figure 5.** Cancer hotspot mutation analysis: delta scores and wild-type/mutant pairs for 7 epitope-altering mutations.

**Figure 6.** t-SNE of the Deep FFN 45-dimensional penultimate-layer representation, coloured by binding class (SB/WB/NB).

**Figure 7.** ESM-2 embedding comparison across five encoding strategies.

**Figure 8.** Rebalanced pan model non-canonical sensitivity (5-seed mean ± SD) across five common HLA-I alleles. Per-allele values are tabulated in Table 6 rather than overlaid, as the per-allele benchmark (n=25 broad-motif) is not directly comparable to the large non-canonical set.

**Figure 9.** Predictor plateau: canonical versus Type B (non-canonical) sensitivity for Deep FFN, MHCflurry, MUNIS, and BigMHC (sequence); dashed line marks the ~0.65 Type B plateau.

**Figure 10.** Pan16: per-allele non-canonical sensitivity for the 15-allele panel, sorted ascending; dashed line marks the pooled 0.819.

**Figure 11.** Temporal holdout: non-canonical sensitivity at threshold 0.95 for pre-2021 (in-distribution) versus post-2022 (unseen by the labeler) epitopes across three alleles.

## Data and code availability

All code, trained models (per-allele `deep_ffn_HLA-A_*.keras` and pan `deep_ffn_pan_seed*.keras`/`deep_ffn_pan16.keras`), feature matrices, prediction results, the curated multi-allele dataset

(`type_ab_dataset.json`, 14,488 positives), and the experimental-label benchmark (6,524 measured IC50 values) are available at github.com/woodhaha/peptide-mhc-benchmark. The per-allele Deep FFN is distributed as the `pmhcbenchmark` Python package, with a `pmhc-predict` command-line tool and a standalone Windows GUI (`pmhc_predict.exe`); exported weights (`pmhc_models.npz`) enable tensorflow-free inference. The single-allele benchmark-hygiene analysis and the multi-allele pan-vs-per-allele benchmark (§3.5–3.11) together constitute the complete study.

## Author contributions

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## Competing interests

The authors declare no competing interests.

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## Ethics statement

This study used only publicly available, de-identified data from the Immune Epitope Database (IEDB) and public protein databases; no human subjects, animals, or patient-derived samples were involved, and no ethics approval or informed consent was required.

## References

1. Falk K, Rotzschke O, Stevanovic S, et al. Allele-specific motifs revealed by sequencing of self-peptides eluted from MHC molecules. Nature 1991;351:290-296.
2. Lundegaard C, Lund O, Nielsen M. State of the art and challenges in sequence based T-cell epitope prediction. Immunome Res 2011;6(Suppl 2):S3.

3. Ruppert J, Sidney J, Celis E, et al. Prominent role of secondary anchor residues in peptide binding to HLA-A2.1 molecules. Cell 1993;74:929-937.
4. Rammensee HG, Friede T, Stevanovic S. MHC ligands and peptide motifs: first listing. Immunogenetics 1995;41:178-228.
5. Reynisson B, Alvarez B, Paul S, et al. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. Nucleic Acids Res 2020;48:W449-W454.
6. O'Donnell TJ, Rubinsteyn A, Laserson U. MHCflurry 2.0: improved pan-allele prediction of MHC class I-presented peptides by incorporating antigen processing. Cell Syst 2020;11:42-48.
7. Albert BA, Yang Y, Shao XM, et al. Deep neural networks predict class I major histocompatibility complex epitope presentation and transfer learn neoepitope immunogenicity. Nat Mach Intell 2023;5:861-872.
8. Wohlwend J, Nathan A, Barzilay R, et al. Deep learning enhances the prediction of HLA class I-presented CD8+ T cell epitopes in foreign pathogens. Nat Mach Intell 2025. doi:10.1038/s42256-024-00971-y.
9. Rives A, Meier J, Sercu T, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proc Natl Acad Sci USA 2021;118:e2016239118.
10. Preibisch G, Tyrolski M, Kucharski P, et al. Resolution of recursive data corruption to transform T-cell epitope discovery. bioRxiv 2026.
11. Henikoff S, Henikoff JG. Amino acid substitution matrices from protein blocks. Proc Natl Acad Sci USA 1992;89:10915-10919.
12. Lin Z, Akin H, Rao R, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science 2023;379:1123-1130.
13. Nielsen M, Lundegaard C, Worning P, et al. Reliable prediction of T-cell epitopes using neural networks with novel sequence representations. Protein Sci 2003;12:1007-1017.
14. Vita R, Mahajan S, Overton JA, et al. The Immune Epitope Database (IEDB): 2018 update. Nucleic Acids Res 2019;47:D339-D343.
15. Malekzadeh P, Yossef R, Cafri G, et al. Antigen experienced T cells from peripheral blood recognize p53 neoantigens. Clin Cancer Res 2020. doi:10.1158/1078-0432.CCR-19-1874.
16. Quandt J, Schlude C, Bartoschek M, et al. Long-peptide vaccination with driver gene mutations in p53 and Kras induces cancer mutation-specific effector as well as regulatory T cell responses. Oncoimmunology 2018;7:e1500671.
17. Wu W, Chen Y, Huang L, et al. Point mutation screening of tumor neoantigens and peptide-induced specific cytotoxic T lym-

phocytes using The Cancer Genome Atlas database. Oncol Lett 2020;20:11986.
18. Hermann L, Fiedler T, Nguyen HA, et al. Beware of data leakage from protein LLM pretraining. Proc Mach Learn Res 2024;261:106-116.
19. Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 2023;4:100804.
20. Bernett J, Blumenthal DB, Grimm DG, et al. Guiding questions to avoid data leakage in biological machine learning applications. Nat Methods 2024. doi:10.1038/s41592-024-02362-y.
21. Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ 2024;385:e078378.
22. Schuh MG, Daniluk A, Sieber SA. Auditing widely used biomolecular benchmarks reveals systematic data inconsistencies. Chem Sci 2026;17:17455-17467.
23. Mares SE, Espinoza Weinberger A, Ioannidis NM. Continued domain-specific pre-training of protein language models for pMHC-I binding prediction. Proc Mach Learn Res 2025;311:304-325.
24. Sarkizova S, Klaeger S, Le PM, et al. A large peptidome dataset improves HLA class I epitope prediction across most of the human population. Nat Biotechnol 2020;38:199-209.
25. Tadros DM, Racle J, Gfeller D. Predicting MHC-I ligands across alleles and species: how far can we go? Genome Med 2025. doi:10.1186/s13073-025-01450-8.
26. Vo TH, McNeela E, O'Donovan O, et al. Artificial intelligence and the evolving landscape of immunopeptidomics. Proteomics Clin Appl 2025. doi:10.1002/prca.70018.
27. Xu H, Ni Y, Chai Z, Cui X, Lei X, Liu L, Shan J, Qian C. TransBindpMHCI: a transformer-based model for pan-specific MHC-I peptide binding prediction. BMC Bioinformatics 2026;27:117.
28. Cheng X, Mei H, Chen P, Wu H, Liu R, Lei Y, Wang P. ESMpHLA: evolutionary scale model-based deep learning prediction of HLA class I binding peptides. HLA 2025;105:e70263.
29. Harndahl M, Rasmussen M, Roder G, et al. Peptide-MHC class I stability is a better predictor than peptide affinity of CTL immunogenicity. Eur J Immunol 2012;42:1405-1416.
30. Backert L, Kohlbacher O. Immunoinformatics and epitope prediction in the age of genomic medicine. Genome Med 2015;7:119.
31. Jappe EC, Garde C, Ramarathinam SH, et al. Thermostability profiling of MHC-bound peptides: a new dimension in immunopeptidomics and aid for immunotherapy design. Nat Commun 2020. doi:10.1038/s41467-020-20166-4.
32. Fasoulis R, Rigo MM, Antunes DA, et al. Transfer learning im-

proves pMHC kinetic stability and immunogenicity predictions. ImmunoInformatics 2024. doi:10.1016/j.immuno.2023.100030.